

AI Smart Grid Monitoring System Report

This report analyzes electricity consumption trends over a 10-year period. The objective is to identify peak demand periods and seasonal patterns using time-series analysis. The results show that electricity demand consistently peaks during summer months due to increased cooling usage.

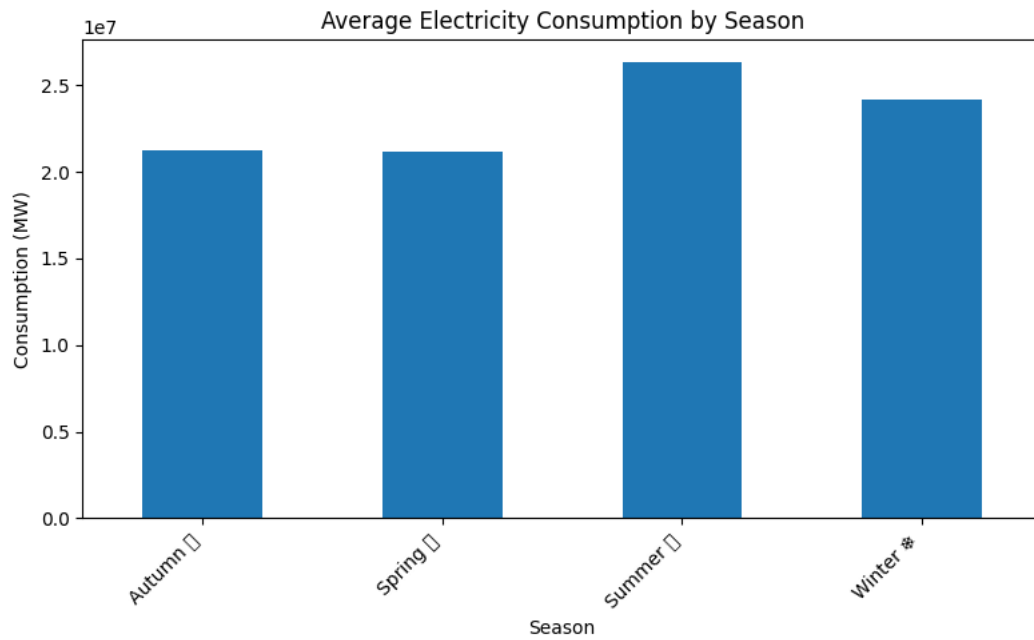
Peak Electricity Consumption by Year

Year	Month	Season	Total Consumption (MW)	Rank (Yearly)	% Change	Remark
2008	July	Summer ■	29,230,849	1	10.63	■ Peak Month
2009	August	Summer ■	27,921,912	1	8.76	■ Peak Month
2010	July	Summer ■	30,201,895	1	12.83	■ Peak Month
2011	July	Summer ■	30,208,932	1	19.77	■ Peak Month
2012	July	Summer ■	29,349,043	1	23.13	■ Peak Month
2013	July	Summer ■	29,153,954	1	21.36	■ Peak Month
2014	January	Winter ❄️■	27,296,284	1	nan	■ Peak Month
2015	July	Summer ■	27,109,704	1	13.34	■ Peak Month
2016	August	Summer ■	28,970,495	1	2.23	■ Peak Month
2017	July	Summer ■	27,151,776	1	14.77	■ Peak Month
2018	July	Summer ■	27,302,373	1	18.45	■ Peak Month

Seasonal Consumption Summary

Season	Total Consumption (MW)
Autumn ■	21,285,041
Spring ■	21,212,164
Summer ■	26,318,590
Winter ❄️■	24,165,679

Seasonal Consumption Visualization



Conclusion: The analysis confirms that electricity consumption is highest during summer months. This is due to increased use of cooling systems such as air conditioning. The findings demonstrate the effectiveness of AI-based time-series analysis in understanding energy demand patterns.